

A Faithful Normalized Cocycle Action Without the Usual Podleś Density Property

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Status: full counterexample to the direct density analogue. There is a normalized faithful cocycle action of the finite discrete quantum group dual to S_3 on $M_2(\mathbb{C})$ for which

$$\overline{\text{span}} \alpha(\mathcal{B})(\mathcal{A} \otimes 1) \neq \mathcal{A} \otimes \mathcal{B}.$$

Thus the density condition equivalent to faithfulness for ordinary actions in Lemma 2.2 of the source is not equivalent to faithfulness for cocycle actions. An exact stabilizer-twisted replacement is also proved below.

1 The source question and precise scope

Adam and Zacharias [1], immediately after Definition 5.1 on arXiv PDF pages 15–16, state the counit characterization of faithfulness for a cocycle action (α, U) and ask whether it is equivalent to “some natural density conditions on the image of α ,” referring to their ordinary-action Lemma 2.2.

where $V = (\epsilon \otimes \epsilon \otimes \text{id}_{\mathcal{B}})(U)$ is a unitary in $\mathcal{B}$ and $\text{Ad}_V(b) = V^*bV$ for all $b \in \mathcal{B}$. We do not know if this property is in turn equivalent to some natural density conditions on the image of α (see Lemma 2.2 for such statement for the non-twisted actions).

If $\mathcal{A} = C_0(G)$ for a classical locally compact group G then $U \in C_b^{\text{strict}}(G \times G; \mathcal{UM}(\mathcal{B}))$, where $\mathcal{UM}(\mathcal{B})$ denotes the group of unitary elements in $\mathcal{M}(\mathcal{B})$. The conditions (5.1) and (5.2) express then the standard cocycle conditions for a twisted action of G on $\mathcal{B}$ (as given for example in [PR]), with the possible exception of the condition $U(e, t) = U(t, e) = 1_{\mathcal{M}(\mathcal{B})}$ for all $t \in G$, which corresponds precisely to the unitary V featuring in (5.3) to be equal to 1.

Figure 1: The question at the top of source PDF page 16.

The displayed formula (5.3) in the published paper is typographically incomplete. This causes no ambiguity here: our example is normalized in the strongest standard sense,

$$(\epsilon \otimes \text{id}_{\mathcal{B}})\alpha = \text{id}_{\mathcal{B}}, \quad (\epsilon \otimes \epsilon \otimes \text{id}_{\mathcal{B}})(U) = 1.$$

The phrase “some natural density conditions” is not itself a formal yes–no proposition. The theorem below settles the canonical concrete interpretation suggested by Lemma 2.2: the same

Podleś density equality is *false*. We do not claim to exclude every condition that might be described informally as natural.

2 Result

Let $\mathcal{A} = C^*(S_3)$, equipped with $\Delta(\lambda_g) = \lambda_g \otimes \lambda_g$ and $\varepsilon(\lambda_g) = 1$. As a finite-dimensional C^* -algebra,

$$\mathcal{A} \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}),$$

where the counit is the trivial-representation scalar block.

Theorem 1. *For $\mathcal{B} = M_2(\mathbb{C})$ there is a unitary $U \in \mathcal{A} \otimes \mathcal{A} \otimes \mathcal{B}$ and a unital faithful $*$ -homomorphism $\alpha : \mathcal{B} \rightarrow \mathcal{A} \otimes \mathcal{B}$ satisfying*

$$\begin{aligned} (\text{id}_{\mathcal{A}} \otimes \alpha)\alpha(b) &= U(\Delta \otimes \text{id}_{\mathcal{B}})\alpha(b)U^*, \\ (\text{id}_{\mathcal{A}} \otimes \text{id}_{\mathcal{A}} \otimes \alpha)(U)(\Delta \otimes \text{id}_{\mathcal{A}} \otimes \text{id}_{\mathcal{B}})(U) &= (1 \otimes U)(\text{id}_{\mathcal{A}} \otimes \Delta \otimes \text{id}_{\mathcal{B}})(U), \end{aligned}$$

and the normalization identities above, but

$$\overline{\text{span}} \alpha(\mathcal{B})(\mathcal{A} \otimes 1) \neq \mathcal{A} \otimes \mathcal{B}.$$

In fact, on the M_2 summand of $\mathcal{A}$, the left-hand side has block exactly $M_2 \otimes \mathbb{C}1$ rather than $M_2 \otimes M_2$.

3 Idea of the proof

The ordinary density condition remembers a distinguished first tensor leg. Exterior perturbation by a unitary preserves faithfulness and turns an ordinary action into a cocycle action, but it need not preserve that tensor leg. On the two-dimensional irreducible block of $C^*(S_3)$, choose the tensor flip. It rotates the trivial action $b \mapsto 1 \otimes b$ into $b \mapsto b \otimes 1$. The latter map remains faithful, while multiplication on the right by the first-leg algebra can never create a non-scalar second leg. The cocycle is the explicit coboundary of the flip.

4 Exterior perturbation lemma

We record the calculation so that no cocycle-existence assertion is hidden.

Lemma 1. *Let $(\mathcal{A}, \Delta)$ be a C^* -bialgebra and let $\beta : \mathcal{B} \rightarrow M(\mathcal{A} \otimes \mathcal{B})$ be an ordinary action. For a unitary $X \in M(\mathcal{A} \otimes \mathcal{B})$ put*

$$\alpha(b) = X^* \beta(b) X, \tag{1}$$

$$U = X_{23}^* (\text{id}_{\mathcal{A}} \otimes \beta)(X^*) (\Delta \otimes \text{id}_{\mathcal{B}})(X). \tag{2}$$

Then (α, U) satisfies both cocycle-action equations. If $\mathcal{A}$ has a counit, β is counital, and $(\varepsilon \otimes \text{id}_{\mathcal{B}})(X) = 1$, then $(\varepsilon \otimes \text{id}_{\mathcal{B}})\alpha = \text{id}_{\mathcal{B}}$ and $(\varepsilon \otimes \varepsilon \otimes \text{id}_{\mathcal{B}})(U) = 1$.

Proof. Coassociativity of β gives $(\text{id} \otimes \beta)\beta = (\Delta \otimes \text{id})\beta$. Using

$$(\text{id} \otimes \alpha)(X^*) = X_{23}^* (\text{id} \otimes \beta)(X^*) X_{23},$$

one obtains

$$(\text{id} \otimes \alpha)\alpha(b) = X_{23}^*(\text{id} \otimes \beta)(X^*)(\Delta \otimes \text{id})\beta(b)(\text{id} \otimes \beta)(X)X_{23}.$$

Inserting (2) and $(\Delta \otimes \text{id})\alpha(b) = (\Delta \otimes \text{id})(X^*)(\Delta \otimes \text{id})\beta(b)(\Delta \otimes \text{id})(X)$ gives the same expression, proving the first cocycle equation.

For completeness, write $q = \text{id}_{\mathcal{A}} \otimes \beta$ and $d = \Delta \otimes \text{id}_{\mathcal{B}}$. Expanding the left side of the second cocycle equation from (2), cancelling adjacent unitary factors, and using $q\beta = d\beta$ reduces it to

$$X_{34}^*q(X^*)_{234}(\text{id} \otimes \Delta \otimes \text{id})q(X^*)(\Delta \otimes \text{id} \otimes \text{id})d(X).$$

The right side reduces to the same expression, with its last factor initially $(\text{id} \otimes \Delta \otimes \text{id})d(X)$; these agree by coassociativity of Δ . This proves the cocycle identity. The final two assertions follow by applying the counit to (1) and (2). $\square$

5 Proof of the counterexample

Take the trivial ordinary action

$$\beta(b) = 1_{\mathcal{A}} \otimes b \quad (b \in \mathcal{B}).$$

Let p be the central projection of the M_2 summand of $\mathcal{A}$, identify $p\mathcal{A} \otimes \mathcal{B}$ with $M_2 \otimes M_2$, and let $F(\xi \otimes \eta) = \eta \otimes \xi$ be the tensor-flip unitary. Define $X \in \mathcal{A} \otimes \mathcal{B}$ blockwise to be 1 on both scalar summands and F on the M_2 summand. In particular, $(\varepsilon \otimes \text{id})(X) = 1$.

Define α and U by (1)–(2). Since β is an action, Lemma 1 proves both cocycle identities and both normalization identities. The map α is faithful because it is a unitary conjugate of the faithful map β .

On the distinguished block,

$$p\alpha(b) = F^*(1_{M_2} \otimes b)F = b \otimes 1_{M_2}.$$

Consequently, if $\mathcal{D} = \overline{\text{span}} \alpha(\mathcal{B})(\mathcal{A} \otimes 1)$, then

$$p\mathcal{D} = \text{span}\{(b \otimes 1)(a \otimes 1) : a, b \in M_2\} = M_2 \otimes \mathbb{C}1,$$

which is a proper subspace of $p(\mathcal{A} \otimes \mathcal{B}) = M_2 \otimes M_2$. This proves Theorem 1.

6 Upgrade: the exact stabilizer-twisted density condition

The example also identifies the missing term in the density condition.

Proposition 1. *Let (α, U) be a cocycle action of a discrete quantum group on $\mathcal{B}$, and suppose a unitary $X \in M(\mathcal{A} \otimes \mathcal{B})$ stabilizes it in the sense that*

$$\beta(b) = X\alpha(b)X^*$$

is an ordinary action. Then

$$\alpha \text{ is faithful} \iff \overline{\text{span}}\{\alpha(b)X^*(a \otimes 1) : a \in \mathcal{A}, b \in \mathcal{B}\} = \mathcal{A} \otimes \mathcal{B}. \quad (\dagger)$$

Proof. Unitary conjugation shows that α is faithful exactly when β is. The ordinary-action Lemma 2.2 of the source says this is equivalent to

$$\overline{\text{span}} X\alpha(\mathcal{B})X^*(\mathcal{A} \otimes 1) = \mathcal{A} \otimes \mathcal{B}.$$

Left multiplication by X^* is a bijective isometry of $\mathcal{A} \otimes \mathcal{B}$ and turns the last display exactly into $(\dagger)$. $\square$

For the flip example, $(\dagger)$ does hold: on the M_2 block,

$$(b \otimes 1)F(a \otimes 1) = F(a \otimes b),$$

whose span is all of $M_2 \otimes M_2$. Thus the failure of the untwisted condition is not an accident; it is exactly repaired by the stabilizer. Whenever X preserves the first-leg module (for example under an appropriate normalizer hypothesis), $(\dagger)$ reduces to the usual condition.

7 Verification and novelty boundary

All identities are finite-dimensional, so there are no strict-topology or one-legged-multiplier issues. The decisive checks are: the exterior perturbation formula, counitality of X , the flip identity, and the strict dimension drop $\dim(M_2 \otimes \mathbb{C}1) = 4 < 16 = \dim(M_2 \otimes M_2)$.

The run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id, title, and the terms “cocycle action,” “density condition,” “Podleś,” “faithful,” and “exterior equivalence.” Targeted searches of later literature located no paper resolving the quoted question or this finite-dimensional construction. The new claim is deliberately limited to the canonical Lemma 2.2 analogue and Proposition 1; the source’s broader informal wording is not overinterpreted.

References

- [1] S. Adam and J. Zacharias, *Approximation properties and entropy estimates for crossed products by actions of amenable discrete quantum groups*, J. London Math. Soc. (2) **82** (2010), 272–288; arXiv:0807.2827.